



Activity 3 — Make Your Tasks Permanent

Time: ~20 minutes · **Files:** `db.js` (new) + `App.js` · **Goal:** your tasks **survive a full restart** using a real SQLite database.

You'll practice today's ideas: `expo-sqlite`, **CRUD** (INSERT/SELECT/UPDATE/DELETE), `useEffect`, and the **refresh** pattern.



Start it up

```
git checkout session-3
npm install          # installs expo-sqlite too
npx expo start
```

If you're upgrading your own Session 2 app instead, first run:

```
npx expo install expo-sqlite
```

Build it in 4 moves

1 Create `db.js` (all the database code)

```
import * as SQLite from 'expo-sqlite';
const db = SQLite.openDatabaseSync('tasks.db');

export function setupDatabase() {
  db.execSync(`CREATE TABLE IF NOT EXISTS tasks (
    id INTEGER PRIMARY KEY NOT NULL,
    title TEXT NOT NULL,
    done INTEGER NOT NULL DEFAULT 0
  );`);
}
export function getTasks() { return db.getAllSync('SELECT * FROM tasks ORDER BY id DE');
export function addTask(title) { db.runSync('INSERT INTO tasks (title, done) VALUES (?, ?');
export function deleteTask(id) { db.runSync('DELETE FROM tasks WHERE id = ?', id); }
```

2 Load saved tasks when the app opens

```
import { useEffect, useState } from 'react';
import * as DB from './db';

const [tasks, setTasks] = useState([]);

useEffect(() => {
  DB.setupDatabase(); // make the table (first run)
  refresh();          // load saved tasks
}, []);               // [] = run once

function refresh() {
  setTasks(DB.getTasks());
}
```

3 Add → save to DB → refresh

```
function add(title) {
  if (title.trim() === '') return;
  DB.addTask(title.trim()); // 1) change the database
  refresh();                // 2) re-read so the screen updates
}
```

4 Delete → remove from DB → refresh

```
function remove(id) {  
  DB.deleteTask(id);  
  refresh();  
}
```

Wire `add` to your input/button and `remove` to each row's 🗑️ (from Session 2).

The big test

1. Add a few tasks.
 2. **Fully close the app** (swipe it away — don't just minimize).
 3. Reopen it.
 4. **Your tasks are still there.** 🎉 You built a real database app!
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✅ Checkpoints — you're done when...

- ☐ `db.js` exists with `setupDatabase`, `getTasks`, `addTask`, `deleteTask`
 - ☐ Tasks **load on open** (via `useEffect`)
 - ☐ **Adding** a task saves it and shows it
 - ☐ **Deleting** a task removes it
 - ☐ Tasks **survive a full restart** ✅✅✅
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🌟 Stretch goals (these prep your final project!)

1. **Toggle done that persists:**

```
export function toggleTask(id, done) {  
  db.runSync('UPDATE tasks SET done = ? WHERE id = ?', done ? 1 : 0, id);  
}
```

2. **Edit a task** with a `Modal` (reuse one form for add + edit — see the slides).
3. **Confirm before delete** with `Alert.alert(...)` .
4. **A floating + button** (`position: 'absolute'`) to open the add form.

The full, polished version of all of these is on the [session-3](#) branch — peek if you get stuck!

😞 Stuck? Quick fixes

Problem	Try this
Cannot find module 'expo-sqlite'	Run <code>npx expo install expo-sqlite</code> , then restart <code>npx expo start</code>
Added a task but the list didn't change	You forgot <code>refresh()</code> after the DB change — call it after every write
Nothing shows on open	Make sure <code>useEffect</code> calls <code>setupDatabase()</code> and <code>refresh()</code>
I changed/deleted everything by accident	Your SQL is missing <code>WHERE id = ?</code>
Old test data won't clear	Delete the app from your phone (this drops the database file) and reopen

🏁 Next: your Final Project

You now have **every skill** to build a real app. Pick a CRUD app you'd actually use — task manager, expense tracker, notes, contacts, habits — and build it.

👉 Full brief, requirements & rubric: [final-project/BRIEF.md](#)